



scrambler:1.6

API Reference

X Platform, Scrambler 2.8.2

12/09/2025

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1 bissCaSettings (1.0)

1.1 Type Reference

1.1.1 BissCaEcmInterval

BISS CA ECM Repetition Interval

Predefined set of possible ECM repetition interval values, in milliseconds.

Values lower than 100 ms are outside of the BISS CA specification. These values may still be used if the parameter `extendedOptions` is set to true in the `BissCaSettings` struct.

enum

I6	5.7 ms (Value outside BISS CA spec)
I15	14.7 ms (Value outside BISS CA spec)
I50	50 ms (Value outside BISS CA spec)
I100	100 ms
I125	125 ms
I150	150 ms
I175	175 ms
I200	200 ms
I250	250 ms
I300	300 ms

1.1.2 BissCaEntitlementFlags

BISS CA Entitlement Flags

The entitlement flags are parameters for how the entitled receivers shall behave. These values are mandatory part of the stream, and is inserted into the session data, which is the encrypted part of the EMM table.

struct

<code>preventDescrambledForward</code>	bool Prevent descrambling forward
<code>preventDecodedForward</code>	bool Prevent decode forward
<code>insertWatermark</code>	bool Insert watermark

1.1.3 BissCaSettings

BISS CA Specific Settings

struct

<code>cryptoPeriod</code>	int Crypto period (in seconds).
<code>emmBitrate</code>	int EMM bitrate (kbps).
	BissCaEcmInterval

ecmInterval	ECM repetition interval. Note that changes to the repetition interval will not take effect until the next crypto period (the next time the ECM content is updated).
emmPreferredPid	int EMM preferred PID value. If the PID value is not available, the first larger PID value available will be used instead. If no larger PID value is available, then the first smaller available value will be used.
ecmPreferredPid	int ECM preferred PID value. If the PID value is not available, the first larger PID value available will be used instead. If no larger PID value is available, then the first smaller available value will be used.
entitlementFlags	BissCaEntitlementFlags BISS CA entitlement flags. Controls how the entitled receivers shall behave.
extendedOptions	bool Allow for extended options. This means allowing violation of the BISS CA specification. If set to true, then the following requirements of the BISS CA specification are <i>not</i> enforced: - EMM repetition interval must be at least 200 ms - ECM repetition interval must be at least 100 ms Note that the requirement that EMM interval must be larger than the ECM interval is still enforced.

2 bissHelpers (1.0)

3 bissScrambling (1.6)

3.1 Type Reference

3.1.1 ScramblerSettings

- Scrambler Settings
-
- Settings defining how to scramble the transport stream.
- By default, there is a single scrambling settings parameter, where these
- settings are applied to all services in the transport stream (TS).
-
- Optionally, however, one or more extra settings may be configured, where the
- different settings are applied to different services in the TS. The different
- settings are referenced by integer indecies. These indecies are
- cross-references to the transport stream configuration. For example, a single
- service may refer to one of these indecies, meaning that the service will be
- scrambled with the corresponding scrambling settings.
-
- Requirements:
 - – All extra scrambling setting *must* contain the same scrambling method as
 - the main settings.
-
- If these requirements are not met, the configuration is not valid.
-
-

struct

enabled	bool Scrambling on/off •
main	ScramblingMethod The main (default) scrambling settings •
extra	map from int to ScramblingMethod Additional scrambling settings, each with its own index.

3.1.2 ScramblingMethod

ScramblingMethod

variant

fixedKeyRaw	bissTypes.FixedKeyRawScrambling Fixed key encryption
biss1	bissVersion1.Scrambling BISS version 1 encryption
biss2	bissVersion2.Scrambling BISS version 2 encryption

4 bissSettings (1.0)

4.1 Type Reference

4.1.1 AllowedExtraSettings

Allowed Num of Extra Scrambling Settings

enum

EXTRA_15_ECMS	15 extra settings
EXTRA_7_ECMS	7 extra settings
EXTRA_3_ECMS	3 extra settings

4.1.2 AllowedMode

Allowed (De)Scrambling modes

enum

ALL	CSA and AES algorithms allowed
AES_ONLY	Only AES allowed
CSA_ONLY	Only CSA allowed

4.1.3 BissModuleSettings

BISS Module Setup for a given product (not configurable)

struct

allowedMode	AllowedMode Algorithms supported by product
numExtraPerOutput	AllowedExtraSettings Number of extra settings allowed per output
maxNumOutputs	int Maximum number of outputs
totalEmmBitrateLimit	optional int Set an upper limit for the total EMM playout bitrate (sum of the bitrate all EMMs). Given in units of kbps.

5 bisstypes (1.3)

5.1 Type Reference

5.1.1 Algorithm

De/Scrambling Algorithm

The DVB-CSA algorithm uses a 64-bit key, while all the AES algorithms use a 128-bit key.

enum

DVB_CSA	DVB-CSA
AES_ATIS_IDSA	AES ATIS IDSA
AES_ECBT_ECB1	AES-ECBT/AES-ECB1 (Alticast)
AES_ECBL_ECB2	AES-ECBL/AES-ECB2 (Latens)
AES_CBCT_CBC1	AES CBCT CBC1
AES_DVB_CISSA	AES DVB-CISSA
AES_IRDETO_CBCT	AES CBCT Irdeto
AES_CBCT_IV	AES CBCT w/specified Initialization Vector

5.1.2 FixedKeyRawDescrambling

Descrambling Method: Fixed Key Raw

struct

algorithm	Algorithm Scrambling algorithm
key	string Scrambling key
iv	optional string Scrambling Initialization vector (only for AES_CBCT_IV)
serviceMappings	list of FixedKeyServiceMapping A list of service mappings.

5.1.3 FixedKeyRawScrambling

Scrambling Method: Fixed Key Raw

struct

algorithm	Algorithm Scrambling algorithm
key	string Scrambling key
iv	optional string Scrambling Initialization vector (only for AES_CBCT_IV)

5.1.4 FixedKeyServiceMapping

Fixed key descrambling for a list of services

struct

key	string Scrambling key
iv	optional string Scrambling Initialization vector (only for AES_CBCT_IV)
serviceIds	list of int A list of services that the key applies to.

6 bissValidate (1.0)

6.1 Type Reference

6.1.1 FloatRange

Float Range

struct

min	float Minimum value of the range
max	float Maximum value of the range

6.1.2 IntRange

Integer Range

struct

min	int Minimum value of the range
max	int Maximum value of the range

6.1.3 Result

Validation result

struct

result	bool True if config is valid and false otherwise.
msg	optional string An error message, which may be given if the result is invalid.

7 bissVersion1 (1.3)

7.1 Type Reference

7.1.1 CsaKeyExpansion

Method of CSA Key Expansion from 48 to 64 bit

enum

CHECKSUM	Session Word is expanded using checksum.
ZERO	Session Word is expanded using zeros.

7.1.2 Descrambling

BISS version 1 Descrambling

variant

mode1	ModelDescrambling BISS version 1, Mode 1.
modeE	ModeEDescrambling BISS version 1, Mode E.

7.1.3 ModelDescrambling

BISS version 1 Descrambling, Mode 1

struct

expansion	CsaKeyExpansion Expansion method.
sessionWord	string Session Word, 48-bit hex string.
serviceMappings	list of ModelServiceMapping A list of service mappings.

7.1.4 ModelScrambling

BISS version 1 Scrambling, Mode 1

struct

expansion	CsaKeyExpansion Expansion method.
sessionWord	string Session Word, 48-bit hex string.

7.1.5 ModelServiceMapping

BISS version 1 Descrambling, Mode 1 for a list of services

struct

sessionWord	string Session Word, 48-bit hex string.
serviceIds	list of int A list of services that the session word applies to.

7.1.6 ModeEDescrambling

BISS version 1 Descrambling, Mode E

struct

activeId	UUID Active session ID.
expansion	CsaKeyExpansion Expansion method.
encSessionWord	UUID Encrypted Session Word, 128-bit. Reference to BissData UUID.
serviceMappings	list of ModeEServiceMapping A list of service mappings.

7.1.7 ModeEScrambling

BISS version 1 Scrambling, Mode E

struct

expansion	CsaKeyExpansion Expansion method.
sessionWord	UUID Session Word, 128-bit. Reference to BissData UUID.

7.1.8 ModeEServiceMapping

BISS version 1 Descrambling, Mode E for a list of services

struct

activeId	UUID Active session ID. Reference to BissData UUID.
encSessionWord	UUID Encrypted Session Word, 128-bit. Reference to BissData UUID.
serviceIds	list of int A list of services that the session ID and encrypted session word applies to.

7.1.9 Scrambling

BISS version 1 Scrambling

variant

mode1	Mode1Scrambling BISS version 1, Mode 1.
-------	---

modeE	ModeEScrambling BISS version 1, Mode E.
-------	---

8 bissVersion2 (1.6)

8.1 Type Reference

8.1.1 Descrambling

BISS version 2 Descrambling

variant

mode1	ModelDescrambling BISS version 2, Mode 1.
modeE	ModeEDescrambling BISS version 2, Mode E.
modeCA	ModeCADescrambling BISS version 2, Mode CA.

8.1.2 ModelDescrambling

BISS version 2 Descrambling, Mode 1

struct

sessionWord	string Session Word, 128-bit hex string.
serviceMappings	list of ModelServiceMapping A list of service mappings.

8.1.3 ModelScrambling

BISS version 2 Scrambling, Mode 1

struct

sessionWord	string Session Word, 128-bit hex string.
-------------	--

8.1.4 ModelServiceMapping

BISS version 2 Descrambling, Mode 1 for a list of services

struct

sessionWord	string Session Word, 128-bit hex string.
serviceIds	list of int A list of services that the session word applies to.

8.1.5 ModeCADescrambling

BISS version 2 Descrambling, Mode CA

struct

key	optional UUID Private RSA key. If UUID value is given, this references a Biss-Data UUID. If optional is empty, the buried RSA private key is used.
-----	--

8.1.6 ModeCAscrambling

BISS version 2 Scrambling, Mode CA

struct

receivers	UUID Public RSA keys of receivers. Reference to BissData CSV File UUID or BissDataGroup UUID.
esId	string Entitlement session ID (16-bit hex string).
settings	bissCaSettings.BissCaSettings ModeCa settings.

8.1.7 ModeEDescrambling

BISS version 2 Descrambling, Mode E

struct

activeId	UUID Active session ID. Reference to BissData UUID.
encSessionWord	UUID Encrypted Session Word, 128-bit. Reference to BissData UUID.
serviceMappings	list of ModeEServiceMapping A list of service mappings.

8.1.8 ModeEScrambling

BISS version 2 Scrambling, Mode E

struct

sessionWord	UUID Session Word, 128-bit. Reference to BissData UUID.
-------------	---

8.1.9 ModeEServiceMapping

BISS version 2 Descrambling, Mode E for a list of services

struct

activeId	UUID Active session ID. Reference to BissData UUID.
encSessionWord	UUID Encrypted Session Word, 128-bit. Reference to BissData UUID.

serviceIds

list of int

A list of services that the session ID and encrypted session word applies to.

8.1.10 Scrambling

BISS version 2 Scrambling

variant

mode1

Mode1Scrambling

BISS version 2, Mode 1.

modeE

ModeEScrambling

BISS version 2, Mode E.

modeCA

ModeCAscrambling

BISS version 2, Mode CA.

9 firewallTypes (1.0)

9.1 Type Reference

9.1.1 IPAddress

string

9.1.2 PaginatedRequest

Instructs the server to return only a subset of the data found on the server

@param pageSize number of entries returned per request

@param pageNumber what page of the set to returned, determined by the slice [pageSize*pageNumber

struct

pageSize	int
pageNumber	int

9.1.3 PaginationInfo

struct

totalEntries	int
--------------	-----

the total number of entries on the server, before matching

9.1.4 SocketAddress

Tuple with IP address and port

struct

address	IPAddress
	IP address, either IPv4 or IPv6
port	int

9.1.5 UUID

string

10 ipinterface (1.0)

10.1 Command Reference

10.1.1 GetIpInterfaces

- message **GetIpInterfaces.Request**
- message **GetIpInterfaces.Response**

10.1.2 SetIpInterfaces

- message **SetIpInterfaces.Request**
- message **SetIpInterfaces.Response**

10.1.3 DeleteIpInterfaces

- message **DeleteIpInterfaces.Request**
- message **DeleteIpInterfaces.Response**

10.2 Type Reference

10.2.1 DeleteIpInterfaces.Request

struct

ids list of UUID

10.2.2 DeleteIpInterfaces.Response

empty struct

10.2.3 GetIpInterfaces.Request

empty struct

10.2.4 GetIpInterfaces.Response

GetIpInterfacesResponse

10.2.5 GetIpInterfacesResponse

struct

data	map from UUID to IpInterface
info	scramblerTypes.PaginationInfo

10.2.6 IpInterface

IP interface configuration, defining interfaces for the physical ports in the system.

struct

enabled	bool Not in use
physicalPortId	UUID UUID of the PhysicalPort
name	string Not in use
vlan	int Not in use
ipv4	optional IpInterfaceConfig IpInterfaceConfig
ipv6	optional IpInterfaceConfig Not in use
deJitterBufferSize	int Not in use
tx	bool Not in use
rx	bool Not in use

10.2.7 IpInterfaceConfig

IP Interface configuration

struct

ipAddress	scramblerTypes.IpAddress IP address
gateway	scramblerTypes.IpAddress Gateway address of the interface
netmask	int Netmask of the interface in CIDR subnet mask notation ie /24 = 255.255.255.0

10.2.8 SetIpInterfaces.Request**struct**

data	map from UUID to IpInterface
------	---

10.2.9 SetIpInterfaces.Responseempty **struct**

11 physicalports (1.4)

11.1 Command Reference

11.1.1 GetPhysicalPorts

- message **GetPhysicalPorts.Request**
- message **GetPhysicalPorts.Response**

11.1.2 SetPhysicalPorts

- message **SetPhysicalPorts.Request**
- message **SetPhysicalPorts.Response**

11.1.3 GetVirtualPorts

- message **GetVirtualPorts.Request**
- message **GetVirtualPorts.Response**

11.1.4 SetVirtualPorts

- message **SetVirtualPorts.Request**
- message **SetVirtualPorts.Response**

11.1.5 GetLinkSpeeds

- message **GetLinkSpeeds.Request**
- message **GetLinkSpeeds.Response**

11.2 Type Reference

11.2.1 GetLinkSpeeds.Request

empty **struct**

11.2.2 GetLinkSpeeds.Response

struct

data **map** from **UUID** to **LinkSpeed**

11.2.3 GetPhysicalPorts.Request

empty **struct**

11.2.4 GetPhysicalPorts.Response

GetPhysicalPortsResponse

11.2.5 GetPhysicalPortsResponse

struct

data	map from UUID to PhysicalPort
info	scramblerTypes.PaginationInfo

11.2.6 GetVirtualPorts.Request

empty struct

11.2.7 GetVirtualPorts.Response

GetVirtualPortsResponse

11.2.8 GetVirtualPortsResponse

struct

data	map from UUID to VirtualPort
info	scramblerTypes.PaginationInfo

11.2.9 LinkSpeed

Link Speed

enum

GBPS_1	1 Gbps
GBPS_10	10 Gbps
UNKNOWN	Unknown speed

11.2.10 PhysicalPort

Physical Port

struct

ifname	string Name of IP interface
label	string User defined label for the Physical Port
portMode	PortMode Port Mode
enabled	bool Disabling will hide the 'No Link' alarm, but does not actually disable the port.

11.2.11 PortMode

Port Mode

enum

RJ45	RJ45 connector
SFP	SFP Plus connector
SFP	SFP connector

11.2.12 SetPhysicalPorts.Request**struct**

data	map from UUID to PhysicalPort
------	-------------------------------

11.2.13 SetPhysicalPorts.Response

empty struct

11.2.14 SetVirtualPorts.Request**struct**

data	map from UUID to VirtualPort
------	------------------------------

11.2.15 SetVirtualPorts.Response

empty struct

11.2.16 VirtualPort

Virtual Port Not in use

struct

physicalPortA	UUID
physicalPortB	UUID
label	string
allowSeamless	bool
allowCloned	bool

12 scramblerAc (1.0)

12.1 Command Reference

12.1.1 GetAcs

- message **GetAcs.Request**
- message **GetAcs.Response**

12.1.2 SetAcs

- message **SetAcs.Request**
- message **SetAcs.Response**

12.1.3 DeleteAcs

- message **DeleteAcs.Request**
- message **DeleteAcs.Response**

12.2 Type Reference

12.2.1 AcConfig

Access Criteria (AC) Config. An AC is connected to a single ECMG, and a single AC group.

struct

acGroupId	UUID UUID of AcGroup this AC is part of
ecmgId	UUID UUID of ECMG this AC is connected to
preferredEcmPid	optional int Preferred ECM PID value
minCryptoPeriod	optional int Minimum crypto period
accessCriteriaType	AccessCriteriaType Access criteria input format
accessCriteria	string Access criteria value
privateData	string Private data content

12.2.2 AccessCriteriaType

Format of the Access Criteria.

enum

INT32	
HEX	

12.2.3 DeleteAcs.Request

struct

ids **list of UUID**

12.2.4 DeleteAcs.Response

empty **struct**

12.2.5 GetAcs.Request

empty **struct**

12.2.6 GetAcs.Response

GetAcsResponse

12.2.7 GetAcsResponse

AcConfig data response

struct

data **map** from **UUID** to **AcConfig**
Map of outputs returned by the request.

info **scramblerTypes.PaginationInfo**
Meta information about the data - i.e the total number of items on the server

12.2.8 SetAcs.Request

struct

data **map** from **UUID** to **AcConfig**

12.2.9 SetAcs.Response

empty **struct**

13 scramblerAcGroup (1.6)

13.1 Command Reference

13.1.1 GetAcGroups

- message **GetAcGroups.Request**
- message **GetAcGroups.Response**

13.1.2 SetAcGroups

- message **SetAcGroups.Request**
- message **SetAcGroups.Response**

13.1.3 DeleteAcGroups

- message **DeleteAcGroups.Request**
- message **DeleteAcGroups.Response**

13.2 Type Reference

13.2.1 AcGroupConfig

Access Criteria Group

Scrambling configuration for a group of transport stream services.

struct

label	string User defined label
mode	ScramblingMode Scrambling mode
disableScrambling	bool Disable all scrambling for group

13.2.2 CaSystem

CA System configuration (DVB SimulCrypt)

Configures the access criteria group to use DVB SimulCrypt. Configuration of ECMG(s) are provided by the EcmgConfig type, and will have to be linked this AcGroupConfig using one or more AcConfig configurations.

struct

minCryptoPeriod	optional int Minimum crypto period for the AC group
descriptorLevel	DescriptorPlacementLevel Placement of the CA descriptor

13.2.3 DeleteAcGroups.Request

struct

ids list of UUID

13.2.4 DeleteAcGroups.Response

empty struct

13.2.5 DescriptorPlacementLevel

Placement of the CA descriptor for all services in the AC group.

enum

BOTH	Default mode; Places CA descriptor on both service and component level
SERVICE	Places CA descriptor only on service level
COMPONENT	Places CA descriptor only on component level

13.2.6 GetAcGroups.Request

empty struct

13.2.7 GetAcGroups.Response

GetAcGroupsResponse

13.2.8 GetAcGroupsResponse

struct

data	map from UUID to AcGroupConfig A map of outputs returned by the request.
info	scramblerTypes.PaginationInfo Meta information about the data - i.e the total number of items on the server

13.2.9 ScramblingMode

Scrambling Mode

What mode is used to scramble the content of this access criteria group.

variant

caSystem	CaSystem CA system (DVB SimulCrypt)
biss	bissScrambling.ScramblingMethod BISS (Basic Interoperable Scrambling System)

13.2.10 SetAcGroups.Request

struct

data **map** from **UUID** to **AcGroupConfig**

13.2.11 SetAcGroups.Response

empty **struct**

14 scramblerAlgorithm (1.3)

14.1 Type Reference

14.1.1 ScramblingAlgorithm

Available scrambling algorithms

enum

DVB_CSA_REDUCED

DVB_CSA_FULL

AES_ATIS_IDSA

AES_ECBT_ECB1

AES_ECBL_ECB2

AES_CBCT_CBC1

AES_DVB_CISSA

AES_IRDETO_CBCT

AES_CBCT_IV

15 scramblerEcmg (1.3)

15.1 Command Reference

15.1.1 GetEcmgs

- message **GetEcmgs.Request**
- message **GetEcmgs.Response**

15.1.2 SetEcmgs

- message **SetEcmgs.Request**
- message **SetEcmgs.Response**

15.1.3 DeleteEcmgs

- message **DeleteEcmgs.Request**
- message **DeleteEcmgs.Response**

15.2 Type Reference

15.2.1 DeleteEcmgs.Request

struct

ids **list of UUID**

15.2.2 DeleteEcmgs.Response

empty **struct**

15.2.3 EcmgConfig

ECM Generator Configuration

struct

label	string User defined label of the ECM generator
algorithm	scramblerAlgorithm.ScramblingAlgorithm Scrambling algorithm used to scramble services that use this ECMG.
casId	int CA system ID allocated by DVB to identify copy protection systems. (uint16)
channelId	int ID of the ECM Simulcrypt channel (uint16)
main	EcmgUnit Main ECMG
redundancyMode	RedundancyMode Redundancy mode for ECM Generator

spare

optional EcmgUnit
Spare ECMG

15.2.4 EcmgUnit

Main/Spare ECM Generator unit.

struct

subCasId

int

CA subsystem ID in combination with the CA system ID gives Super CA system ID which uniquely identifies the ECMG. (uint16_t)

address

scramblerTypes.SocketAddress
Address/Port of the ECMG

15.2.5 GetEcmgs.Request

empty **struct**

15.2.6 GetEcmgs.Response

GetEcmgsResponse

15.2.7 GetEcmgsResponse

struct

data

map from **UUID** to **EcmgConfig**
A map of outputs returned by the request.

info

scramblerTypes.PaginationInfo
Meta information about the data - i.e the total number of items on the server

15.2.8 RedundancyMode

Redundancy mode for an ECM Generator

enum

DISABLED

COLD

15.2.9 SetEcmgs.Request

struct

data

map from **UUID** to **EcmgConfig**

15.2.10 SetEcmgs.Response

empty **struct**

16 scramblerEis (1.0)

16.1 Command Reference

16.1.1 GetEis

- message **GetEis.Request**
- message **GetEis.Response**

16.1.2 SetEis

- message **SetEis.Request**
- message **SetEis.Response**

16.1.3 DeleteEis

- message **DeleteEis.Request**
- message **DeleteEis.Response**

16.2 Type Reference

16.2.1 DeleteEis.Request

struct

ids **list** of **UUID**

16.2.2 DeleteEis.Response

empty **struct**

16.2.3 EisConfig

EIS Interface configuration

struct

label	string User defined label for EIS Interface
localPort	int Local port where EIS server is expected to send messages
remoteIps	list of scramblerTypes.IpAddress Remote IP addresses where the EIS messages are accepted from

16.2.4 GetEis.Request

empty **struct**

16.2.5 GetEis.Response

GetEisResponse

16.2.6 GetEisResponse

struct

data

map from UUID to EisConfig

A map of outputs returned by the request.

info

scramblerTypes.PaginationInfo

Meta information about the data - i.e the total number of items on the server

16.2.7 SetEis.Request

struct

data

map from UUID to EisConfig

16.2.8 SetEis.Response

empty struct

17 scramblerEmm (1.0)

17.1 Command Reference

17.1.1 GetEmms

- message **GetEmms.Request**
- message **GetEmms.Response**

17.1.2 SetEmms

- message **SetEmms.Request**
- message **SetEmms.Response**

17.1.3 DeleteEmms

- message **DeleteEmms.Request**
- message **DeleteEmms.Response**

17.2 Type Reference

17.2.1 DeleteEmms.Request

struct

ids **list of UUID**

17.2.2 DeleteEmms.Response

empty **struct**

17.2.3 EmmConfig

EMM Interface

struct

label	string User defined label of the EMM Stream
emmgId	UUID UUID of the EMM Generator
streamId	int EMM Stream ID (uint16)
udpPort	optional int Local UDP listening port. Not applicable for EMMG connection type TCP.
maxBitrate	int Maximum accepted EMM bitrate (in kbps)
emmType	EmmType EMM type
	optional string

privateData	Private Data for CA descriptor in the CAT, hex string max 251 bytes. Not applicable for EMM Type PRIVATE_DATA.
pid	optional int Preferred EMM PID

17.2.4 EmmType

EMM Type

enum

EMM	EMM is signaled in the CAT
PRIVATE_DATA	Unsignaled data

17.2.5 GetEmms.Request

empty **struct**

17.2.6 GetEmms.Response

GetEmmsResponse

17.2.7 GetEmmsResponse

struct

data	map from UUID to EmmConfig A map of outputs returned by the request.
info	scramblerTypes.PaginationInfo Meta information about the data - i.e the total number of items on the server

17.2.8 SetEmms.Request

struct

data	map from UUID to EmmConfig
------	---

17.2.9 SetEmms.Response

empty **struct**

18 scramblerEmmg (1.1)

18.1 Command Reference

18.1.1 GetEmmgs

- message **GetEmmgs.Request**
- message **GetEmmgs.Response**

18.1.2 SetEmmgs

- message **SetEmmgs.Request**
- message **SetEmmgs.Response**

18.1.3 DeleteEmmgs

- message **DeleteEmmgs.Request**
- message **DeleteEmmgs.Response**

18.2 Type Reference

18.2.1 ConnectionType

Connection type

enum

TCP
UDP
TCP_UDP

18.2.2 DeleteEmmgs.Request

struct

ids **list of UUID**

18.2.3 DeleteEmmgs.Response

empty **struct**

18.2.4 EmmgConfig

EMM Generator Configuration

struct

label **string**
User defined label of the EMM Generator
int

casId	CA system ID allocated by DVB to identify copy protection systems. (uint16)
subCasId	int CA subsystem ID in combination with the CA system ID gives Super CA system ID which uniquely identifies the ECMG. (uint16_t)
connectionType	ConnectionType Connection type used towards the EMMG
tcpPort	optional int Local TCP port. Not applicable for UDP connection type.
remoteIps	list of scramblerTypes.IpAddress Remote IP addresses where the messages from EMMG are accepted.

18.2.5 GetEmmgs.Request

empty **struct**

18.2.6 GetEmmgs.Response

GetEmmgsResponse

18.2.7 GetEmmgsResponse

struct

data	map from UUID to EmmgConfig A map of outputs returned by the request.
info	scramblerTypes.PaginationInfo Meta information about the data - i.e the total number of items on the server

18.2.8 SetEmmgs.Request

struct

data	map from UUID to EmmgConfig
------	--

18.2.9 SetEmmgs.Response

empty **struct**

19 scramblerScrService (1.1)

19.1 Command Reference

19.1.1 GetScrServices

- message **GetScrServices.Request**
- message **GetScrServices.Response**

19.1.2 SetScrServices

- message **SetScrServices.Request**
- message **SetScrServices.Response**

19.1.3 DeleteScrServices

- message **DeleteScrServices.Request**
- message **DeleteScrServices.Response**

19.2 Type Reference

19.2.1 ComponentGroups

Component groups, type of PIDs to scramble.

enum

```
AUDIO_VIDEO_TELETEXT
AUDIO_VIDEO
VIDEO
```

19.2.2 DeleteScrServices.Request

struct

ids **list of UUID**

19.2.3 DeleteScrServices.Response

empty **struct**

19.2.4 GetScrServices.Request

empty **struct**

19.2.5 GetScrServices.Response

GetScrServicesResponse

19.2.6 GetScrServicesResponse

struct

data	map from UUID to ScrServiceConfig A map of outputs returned by the request.
info	scramblerTypes.PaginationInfo Meta information about the data - i.e the total number of items on the server

19.2.7 ScrServiceConfig

Scrambling options of each service.

struct

sourceNodeId	UUID UUID of the service source
acGroupId	UUID UUID of the AC Group
components	ComponentGroups Components to be scrambled
partialScrambling	optional int Percentage of packets to be scrambled; [0, 100], 0=clear, 100=full scrambling
pesHeaderInClear	optional int Do not scramble packets with PUSI=1. In addition, the given number of packets, [0, 7] following the PUSI packet (on the same PID) will also not be scrambled.

19.2.8 SetScrServices.Request

struct

data	map from UUID to ScrServiceConfig
------	--

19.2.9 SetScrServices.Response

empty **struct**

20 scramblerStatus (1.3)

20.1 Command Reference

20.1.1 GetEcmgStatus

Get status for given ECMGs

- message **GetEcmgStatus.Request**
- message **GetEcmgStatus.Response**

20.1.2 GetEcmStatus

Get status for given ECMs

- message **GetEcmStatus.Request**
- message **GetEcmStatus.Response**

20.1.3 GetScgStatus

Get status for given SCGs

- message **GetScgStatus.Request**
- message **GetScgStatus.Response**

20.1.4 GetEisStatus

Get status for given EISs

- message **GetEisStatus.Request**
- message **GetEisStatus.Response**

20.1.5 GetEmmgStatus

Get status for given EMMGs

- message **GetEmmgStatus.Request**
- message **GetEmmgStatus.Response**

20.1.6 GetEmmStatus

Get status for given EMMs

- message **GetEmmStatus.Request**
- message **GetEmmStatus.Response**

20.2 Type Reference

20.2.1 EcmStatus

State of ECM

struct

state

State

20.2.2 EcmgStatus

State of ECMG

struct

mainState	State State of Main ECMG
spareState	State State of Spare ECMG

20.2.3 EisStatus

Status of EIS

struct

lastConnectedIp	string The last IP the EIS messages are received from
currentState	State Current status of the EIS

20.2.4 EmmStatus

Status of EMM

struct

rxBytes	optional bigint Number of received bytes
---------	--

20.2.5 EmmgStatus

Status of EMMG

struct

lastConnectedIp	string Remote IP of the last EMMG connection
currentState	State Current status of the EMMG

20.2.6 GetEcmStatus.Request

struct

ids	optional list of UUID
-----	------------------------------

20.2.7 GetEcmStatus.Response

struct

data map from UUID to EcmStatus

20.2.8 GetEcmgStatus.Request

struct

ids optional UuidList

20.2.9 GetEcmgStatus.Response

struct

data map from UUID to EcmgStatus

20.2.10 GetEisStatus.Request

struct

ids optional list of UUID

20.2.11 GetEisStatus.Response

struct

data map from UUID to EisStatus

20.2.12 GetEmmStatus.Request

struct

ids optional list of UUID

20.2.13 GetEmmStatus.Response

struct

data map from UUID to EmmStatus

20.2.14 GetEmmgStatus.Request

struct

ids optional list of UUID

20.2.15 GetEmmgStatus.Response

struct

data map from UUID to EmmgStatus

20.2.16 GetScgStatus.Request

struct

ids optional list of UUID

20.2.17 GetScgStatus.Response

struct

data map from UUID to ScgStatus

20.2.18 ScgStatus

State of SCG

struct

cryptoPeriodNumber	optional int Crypto Period Number
activeAlgorithm	optional scramblerAlgorithm.ScramblingAlgorithm Scrambling algorithm

20.2.19 State

State

enum

OK
WARNING
ERROR
NEUTRAL
UNKNOWN

20.2.20 UuidList

list of UUID

21 scramblerTypes (1.0)

21.1 Type Reference

21.1.1 IPAddress

string

21.1.2 PID

int

21.1.3 PaginatedRequest

Instructs the server to return only a subset of the data found on the server.

struct

pageSize	int Number of entries returned per request
pageNumber	int Page of the set to returned, determined by the slice [page-Size*pageNumber : (pageSize+1)*pageNumber]

21.1.4 PaginationInfo

struct

totalEntries	int Total number of entries on the server, before matching
--------------	---

21.1.5 ScramblingAlgorithm

Available scrambling algorithms

enum

```
DVB_CSA_V1
DVB_CSA_V2
AES_ATIS_IDSA
AES_ECBT_ECB1
AES_ECBL_ECB2
AES_CBCT_CBC1
AES_DVB_CISSA
AES_IRDETO_CBCT
AES_CBCT_IV
```

21.1.6 SocketAddress

Tuple with IP address and port

struct

ip	IpAddress IP address, either IPv4 or IPv6
port	int

21.1.7 UUID**string**

22 scramblerValidate (1.0)

22.1 Type Reference

22.1.1 Range

struct

min	int
max	int

22.1.2 Result

struct

result	bool
msg	optional string

23 services (1.0)

23.1 Command Reference

23.1.1 GetInputServices

Returns service information for the queried list of service node IDs

- message **GetInputServices.Request**
- message **GetInputServices.Response**

23.1.2 GetServiceDependencies

Returns service dependency information

- message **GetServiceDependencies.Request**
- message **GetServiceDependencies.Response**

23.2 Type Reference

23.2.1 ChildSourceInfo

struct

name	string
nodeType	string
body	string
owner	string

23.2.2 FlowSourceInfo

struct

name	string
label	string
nodeType	string
body	string
sources	map from UUID to ChildSourceInfo
owner	string

23.2.3 GetInputServices.Request

empty **struct**

23.2.4 GetInputServices.Response

struct

data	map from UUID to FlowSourceInfo
------	--

23.2.5 GetServiceDependencies.Request

empty **struct**

23.2.6 GetServiceDependencies.Response

struct

data **list** of **ServiceDependencyInfo**

23.2.7 ServiceDependencyInfo

struct

source	UUID
dependency	UUID

23.2.8 ServiceSourceInfo

struct

serviceId	int
name	string
owner	string